Life Processes in living organisms Part -1

I] Textbook Questions:

1 Mark Objectives:

1. Write definitions.

a. **Nutrition:** The process of intake of food and its utilization by an organism is called as nutrition.
b. **Nutrients**: Nutrients are components of food which provide energy and help in growth and maintenance of the body.
c. **Proteins**: Proteins are the macromolecules formed by bonding together many amino acids.
d. **Cellular respiration**: Cellular respiration is a process where foodstuffs are oxidised either with or without the help of oxygen to produce energy.
e. **Aerobic respiration**: Aerobic respiration is the process of producing energy from foodstuffs with the help of oxygen.
f. **Glycolysis**: Glycolysis is the process in which a molecule of glucose is oxidized in a step-by-step process and two molecules of each, i.e. pyruvic acid, ATP, $NADH_2$ and water are formed.

Q.2. Fill in the blanks and explain the statements.

1. After complete oxidation of a glucose molecule, **38** number of ATP molecules are formed. *[2026]*
2. At the end of glycolysis, **pyruvic acid, ATP, $NADH_2$, and water** molecules are obtained.
3. Genetic recombination occurs in the **4th phase- Pachytene** of prophase of meiosis-I.
4. All chromosomes are arranged parallel to the equatorial plane of a cell in the **metaphase** of mitosis. *[2021]*
5. For the formation of plasma membrane, **phospholipid** molecules are necessary.
6. Our muscle cells perform **anaerobic** type of respiration during exercise. *[2025]*

2 Marks Questions:

Q. Give scientific reasons:

1. Oxygen is necessary for complete oxidation of glucose.
Ans:

1. In living organisms, cellular respiration occurs by two methods: aerobic respiration and anaerobic respiration.
2. Aerobic cellular respiration is the process by which cells use oxygen to help them convert glucose into energy.
3. In Anaerobic respiration, without oxygen, glucose will be incompletely oxidized and very less energy (2 ATP) is produced.

4. In Aerobic respiration, in the presence of oxygen, glucose is completely oxidised to produce more energy (38 ATP). Therefore, oxygen is necessary for complete oxidation of glucose.

2. Fibers are one of the important nutrients. *[2024, 2025]*

Ans:

1. Along with all other nutrients, fibers are also essential nutrients.
2. In fact, we cannot digest the fibers.
3. However, they help in the digestion of other substances and egestion of undigested substances.
4. We obtain the fibers from leafy vegetables, fruits, cereals, etc.

3. Cell division is one of the important properties of cells and organisms. *[2021, 2022]*

Ans:

1. Cell division is one of the very important properties of cells and living organisms.
2. Due to this property only, a new organism is formed from existing one, a multicellular organism grows up and emaciated body can be restored.
3. There are two types of cell division as mitosis and meiosis.
4. Mitosis occurs in somatic cells and stem cells of the body whereas meiosis occurs in germ cells.

4. Sometimes, higher plants and animals too perform anaerobic respiration. *[2023, 2025]*

OR **We feel tired after exercise.** *[2024, 2020, 2020]*

Ans:

1. Some higher plants, animals and aerobic microorganisms also perform anaerobic respiration instead of aerobic respiration if there is depletion in oxygen level in the surrounding.
2. E.g. Seeds perform anaerobic respiration if the soil is submerged under water during germination.
3. Similarly, our muscle cells also perform anaerobic respiration while performing the exercise.
4. Due to this, less amount of energy is produced in our body and lactic acid accumulates due to which we feel tired.

5. Kreb's cycle is also known as citric acid cycle.

Ans:

1. In Kreb’s cycle, both molecules of acetyl-CoA enter the mitochondria.
2. Cyclic chain of reactions called as tricarboxylic acid cycle is operated on it in the mitochondria.
3. Acetyl part of acetyl-CoA is completely oxidized through this cyclical process and molecules CO_2 , H_2O, $NADH_2$, $FADH_2$ are derived.
4. Since, citric acid is the very first stable product generated, Kreb’s cycle is also known as citric acid cycle.

Q. Distinguish between:

** Two distinguishing points for two marks and three points for three marks.*

1. Mitosis and Meiosis. *[2021]*

Ans:

Mitosis	Meiosis
1. Mitosis occurs in somatic cells and stem cells.	1. Meiosis occurs in germ cells.
2. The daughter cells formed, maintain the original chromosome number.	2. The number of chromosomes become half (2n to n) in the daughter cells.
3. The daughter cells are diploid (2n).	3. The daughter cells are haploid (n).
4. In this division 2 new cells are formed.	4. In this division 4 new cells are formed.
5. It is important for growth and restoration of emaciated body.	5. This division is required for production of gametes.

2. Glycolysis and TCA cycle.

Ans:

Glycolysis	TCA cycle
(i) Glycolysis occurs in the cytoplasm.	(i) TCA cycle occurs in the mitochondria.
(ii) Two molecules each of pyruvic acid, ATP, $NADH_2$ and water are formed.	(ii) Molecules of CO_2, H_2O, $NADH_2$ and $FADH_2$ are formed.
(iii) It is also called as EMP pathway.	(iii) It is also called as Kreb's cycle.
(iv) No carbon dioxide is released.	(iv) Carbon dioxide is released.

3. Aerobic and anaerobic respiration. *[2020]*

Ans:

Aerobic respiration	Anaerobic respiration
(i) It occurs in the presence of oxygen.	(i) It occurs in the absence of oxygen.
(ii) Common in all higher plants and animals.	(ii) Common in certain microorganisms but very rare in all higher plants and animals.

Aerobic respiration	Anaerobic respiration
(iii) Energy is released in greater amounts in the form of ATP.	(iii) Energy is released in lesser amounts in the form of ATP.
(iv) Glucose is completely oxidised.	(iv) Glucose is incompletely oxidised.
(v) The end products are CO_2 and water.	(v) The end products are CO_2 and organic acids or alcohol.

⭐ 3 Marks Questions: ⭐

Q. Answer the following questions.

1. Explain glycolysis in detail.

Ans:

1. Process of glycolysis occurs in cytoplasm.
2. A molecule of glucose is oxidized step by step in this process and two molecules of each i.e. pyruvic acid, ATP, $NADH_2$ and water are formed.
3. Molecules of pyruvic acid formed in this process are converted into molecules of Acetyl-Coenzyme-A.
4. Two molecules of $NADH_2$ and two molecules of CO_2 are released during this process.

2. How all the life processes contribute to the growth and development of the body?

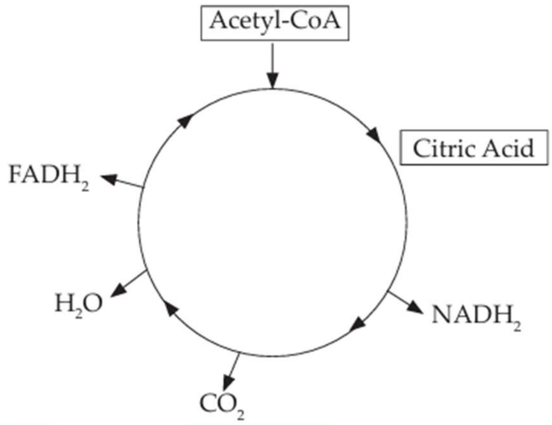
Ans:

1. Various organ systems are continuously performing life processes like digestion, respiration, circulation, and excretion are performed independently but with complete co-ordination.
2. Digestion is the process where complex food is broken down into simple food. The digested food is carried to all the cells, where in the presence of oxygen, energy is produced in each cell i.e. respiration.
3. The waste is expelled out of the body through the process of excretion.
4. The digested food, oxygen, etc., are transferred to different parts of the body through circulation.
5. Reproduction is another life process which helps in the perpetuation of species.
6. Thus, we can see that all the life processes, coordinated with each other, bring about complete growth and development of an organism.

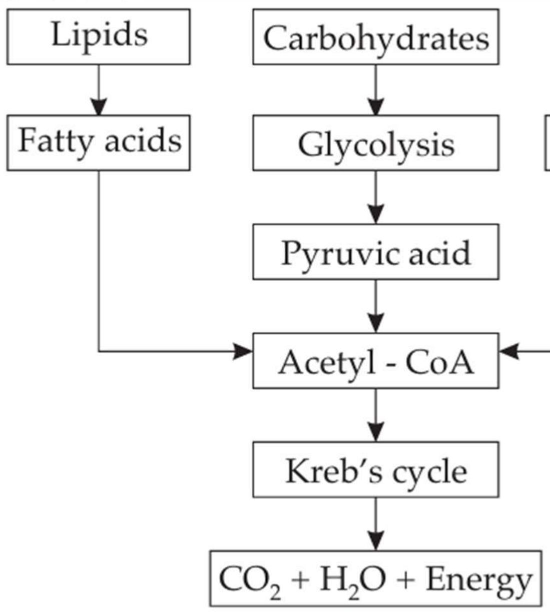
3. Explain the Kreb's cycle with reaction.

Ans:

1. Two molecules of acetyl Co-A produced from pyruvic acid (formed by glycolysis) enter the mitochondria.
2. Cyclic chain of reactions called as tricarboxylic acid cycle is operated on it in the mitochondria.
3. Acetyl part of acetyl-CoA is completely oxidized through this cyclical process and molecules of CO_2 , H_2O, $NADH_2$, $FADH_2$ are formed.



3. How is energy formed from oxidation of carbohydrates, fats and proteins? Correct the diagram given below. *[2024, 2025, 2019]*



Ans:

⭐ 5 Marks Questions: ⭐

Q.1. With the help of suitable diagrams, explain the mitosis in detail.

OR Write any four stages of mitosis and explain any two of them. [3m] *[2025]*

OR Name the type of cells in which mitosis occurs? Write four stages of karyokinesis. [3m] *[2026]*

Ans:

Mitosis: It occurs in somatic cells.

I] Karyokinesis: (Division of Nucleus)

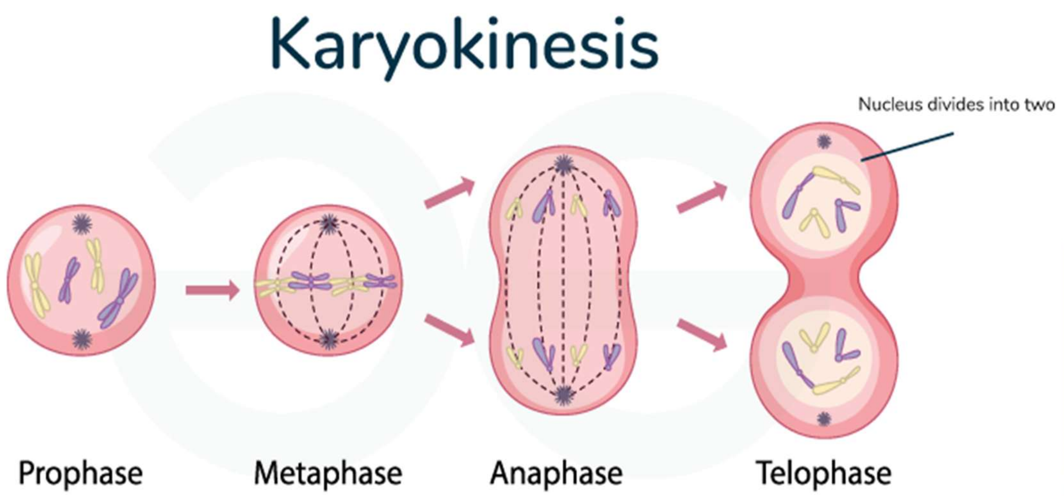
It occurs in 4 phases.

1. **Prophase:**
   i. Chromosomes condense and appear along with sister chromatids.
   ii. Centrioles duplicate and move to the opposite poles of cell.
   iii. Nuclear membrane and nucleolus start disappearing.

2. **Metaphase:**
   i. Nuclear membrane and nucleus disappear.
   ii. Chromosomes arrange themselves to the Equatorial Plain.
   iii. spindle fibres are formed between centromere and centriole.

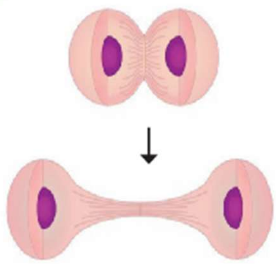
3. **Anaphase:**
   i. Sister chromatids split up as they are pulled up by the spindle fibres in opposite directions/ poles.
   ii. Daughter chromosomes are formed.

4. **Telophase:**
   i. Chromosomes at polls decondense and become thread-like invisible.
   ii. Nuclear membrane is formed at each pole and two daughter nuclei along with nucleolus are formed in the cell.
   iii. Spindle fibres disappear completely.



II] Cytokinesis (Division of cytoplasm):

Notch formed at the Equatorial Plain deepens gradually and two daughter cells are formed after separation.



2.7 Cytokinesis

II] Extra - Previous Years' Questions: (PYQs)

⭐ 1 Mark Objectives: ⭐

MCQS:

1. A molecule of glucose is completely oxidised in aerobic respiration and molecules of H_2O and __________ are produced along with energy. *[2025]*

(A) CO_2 (B) O_2 (C) NaOH (D) HNO_3

2. We get __________ kCal of energy per gram of proteins. *[2024]*

(A) 4 (B) 3 (C) 2 (D) 1

3. We get __________ kCal of energy per gram of lipid. *[2023]*

(A) 9 (B) 4 (C) 2 (D) 1

4. Bones contain __________ amino acid. *[2022]*

(A) Melanin (B) Haemoglobin **(C) Ossein** (D) Insulin

6. **Correlations**: Skin: Melanin :: Pancreas : **Insulin** *[2024]*

7. Odd one out: Actin, Ossein, Myosin, **Aldosterone** *[2023]*

8. True or False: Oxidation of proteins is carried out in aerobic respiration. **(False)** *[2022]*

⭐ 2 Marks Questions: ⭐

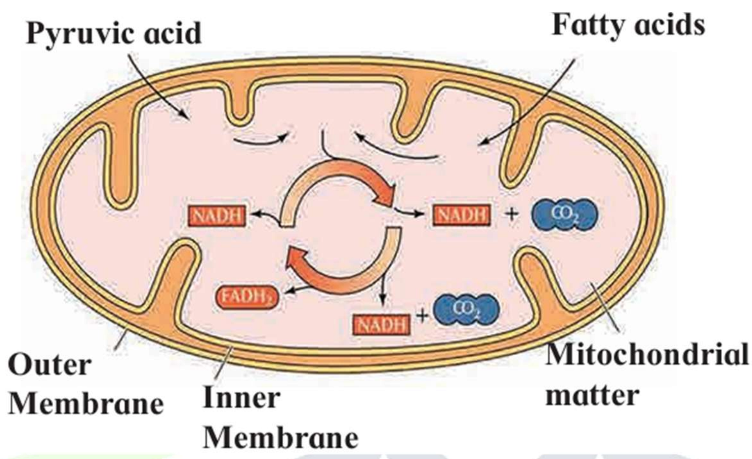
Give Reasons:

1. **Water is an essential nutrient:** *[2022]*

Ans:

1. There is about 65 – 70% water in our body.
2. Each cell contains 70% water weight by weight.
3. Blood-plasma also contains 90% of water.
4. Functioning of cells and thereby whole body disturbs even if there is a little loss of water from the body. Hence, water is an essential nutrient.

2. Draw a well labelled diagram of mitochondria and tri-carboxylic acid cycle. *[2023]*



3. What are vitamins? Write two types of vitamins. *[2022, 2024]*

Ans:

1. Vitamins are a group of heterogeneous compounds of which, each is essential for proper operation of various processes in the body.
2. There is main six types of vitamins, e.g. A, B, C, D, E and K.
3. Out of these, A, D, E and K are fat-soluble whereas B and C are water-soluble.

⭐ 3 Marks Questions: ⭐

1. Complete the flow chart. *[2023, 2026]*

Sr. No.	Proteins	Name of the Organs
1.	Melanin, Keratin	**Skin**
2.	**Insulin, Trypsin**	Pancreas
3.	Haemoglobin	**Blood**
4.	**Ossein**	Bone

2. **1. Write two sources of lipids.** *[2022]*

Ans: Milk, butter, cheese, oil, ghee, meat, nuts are various sources of lipids**.**

2. How much energy is obtained from lipids?

Ans: 9 kCal of energy is obtained from lipids.

3. What is produced after digestion of lipids?

Ans: Fatty acid and alcohol are produced after digestion of lipids**.**